

SALARY 2045

Portfolio Rebalancer — User Guide

Buy-Only · Multi-Currency · v6

This guide explains how the Salary2045 Rebalancer works — from basic investing concepts to the exact formulas used under the hood. Written for anyone, including first-time investors.

Contents

1. What is the Rebalancer?
2. Key Concepts & Terminology
3. The Interface — Fields Explained
4. How the Allocation Engine Works
5. Transaction Fees & Version B
6. Multi-Currency Support
7. Reading the Results
8. Formula Reference
9. Common Scenarios

1. What is the Rebalancer?

The Salary2045 Rebalancer answers one practical question:

"I have X dollars of new cash. How do I invest it to bring my portfolio closer to its target allocation?"

It is **buy-only** — it never tells you to sell anything. It only tells you what to buy with the new money you are adding right now. Over time, as you add money regularly (e.g. monthly salary savings), your portfolio gradually converges to your target allocation without triggering tax events from selling.

Why buy-only?

Selling creates taxable events and brokerage fees. Buying only, gradually, is the low-cost path to a balanced portfolio. The math is simple: direct new money to what is underweight, and over time everything aligns.

Example

You target 40% in stable dividend ETFs and 40% in growth stocks. After a strong month for growth, growth sits at 50% and dividends dropped to 30%. Instead of selling growth, you direct this month's savings entirely toward dividends — they catch up naturally.

2. Key Concepts & Terminology

Portfolio

The total value of all your holdings (stocks, ETFs, bonds) plus your cash balance, all converted to your chosen base currency.

Sleeve (Allocation Bucket)

A named category that groups related holdings. Examples: "Core" for stable ETFs, "Satellite" for individual stocks, "Hedge" for gold. Each sleeve has a target percentage of the total portfolio.

■ *Think of sleeves as buckets. You decide what % of your money belongs in each bucket. The rebalancer fills the emptiest buckets first.*

Target Allocation

The percentage you want each sleeve to represent. All targets must add to exactly 100%.

Core 40% + Satellite 38% + High Conviction 10% + Hedge 7% + Cash 5% = 100%

totalNow vs totalAfter

totalNow	Current portfolio value before adding new cash. = all holdings + existing cash.
totalAfter	Portfolio value AFTER new cash is added. = totalNow + newCash. This is the denominator for ALL target calculations.

■ *All target percentages are calculated against totalAfter — where you want to end up, not where you are now.*

Cash Sleeve

One sleeve is designated as "Cash." It has two thresholds:

- **Hard floor: 1% of totalAfter.** If cash drops below this, it is topped up first before anything else.
- **Soft target:** your set % (e.g. 5%). Cash is topped up here last, only after securities are funded.

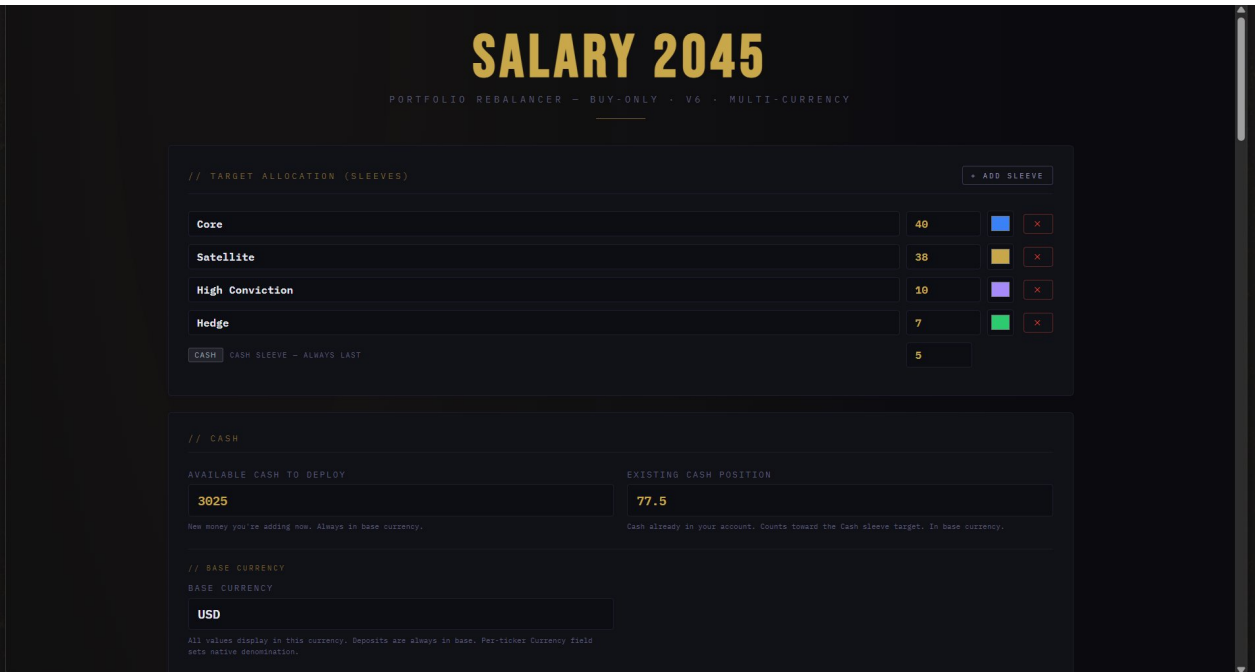
Max % Cap (per ticker)

An optional ceiling for a single stock. If a stock is already at 8% and your cap is 8%, it receives zero new allocation.

Fractionable Shares

Some brokers allow buying 0.37 shares of a stock. If Frac is OFF, the tool only buys whole units and returns the sub-unit remainder to Cash Left.

3. The Interface — Fields Explained



Sleeves panel (top) and Cash inputs.

Sleeve Panel

Name	Label for this group — e.g. "Core" or "Satellite."
Target %	The % of totalAfter you want this sleeve to reach.
Cash sleeve	Marked "CASH SLEEVE — ALWAYS LAST." One sleeve is the cash bucket.

Cash Panel

Available Cash to Deploy	New money you are adding now. Always in base currency.
Existing Cash Position	Cash already in your brokerage account. Counts toward the Cash sleeve target but is NOT reinvested.
Base Currency	The currency all values display in. Set to USD, ILS, EUR, etc.

// TRANSACTION FEES (OPTIONAL)

FIXED FEE PER TRADE

2.5

PERCENTAGE FEE PER TRADE (%)

0

Flat amount (base currency) charged per trade.

% of trade size. Both filled = fee = fixed + pct * trade.

// OPTIMIZATION MODE

TRADE STRATEGY

Version A — Maximize Allocation Accuracy

Version B cuts smallest trades until T fees < budget. Requires fees > 0 — automatically falls back to A when fees are cleared.

// WHOLE-UNIT ROUND-UP TOLERANCE

OVERSHOOT TOLERANCE (%)

1

Max % of portfolio value a whole-unit ticker may overshoot its Phase 1 target when rounding up (only when remainder > 0.5 unit). Set 0 to always round down.

// LIVE PRICES (OPTIONAL)

PROVIDER

Finnhub

API KEY

Paste your key — leave empty for manual entry

Pick your provider, paste its key, then "Refresh Prices". The key stays in your browser.

Fees, optimization mode, and overshoot tolerance.

Fees & Mode Panel

Fixed Fee per Trade	Flat amount charged per buy order, e.g. \$2.50.
Percentage Fee (%)	Fee as % of trade size. Both fixed and % can be combined.
Version A	Accuracy-first. All eligible tickers receive allocation regardless of fee impact.
Version B	Budget-first. Smallest trades are cut until total fees fit within your Max Fee Budget.
Overshoot Tolerance	How much a whole-unit ticker may exceed its target when rounding up to the next full share (only when remainder ≥ 0.5 units). Set 0 to always round down.

// FX RATES (BASE: USD)

REFRESH RATES

Rates from frankfurter.app (free, no key). Editable — override manually if offline or to fix a rate. Only currencies currently in use are shown. Switching base clears rates and auto-fetches.

CURRENCY

1 USD =

USD

BASE

1.000000

Add holdings with non-base currencies to see conversion rows.

Fetch failed: Failed to fetch. Manual rates active.

FX Rates panel — auto-fetched from frankfurter.app, manually editable.

FX Rates Panel

Rates are fetched from frankfurter.app (free, no API key needed). If offline or running from a local file, enter rates manually. Only currencies currently in use are shown.

// HOLDINGS REFRESH PRICES

TICKER	SLEEVE	SHARES	PRICE (NATIVE)	CURR	VALUE (USD)	MAX %	FRAC
SCHD	Core	52	27.7	USD	\$1,449.49	—	☐
DGRW	Core	5	98.39	USD	\$451.95	—	☐
CNI	Satellite	3	100	USD	\$389.89	—	☐
PLD	Satellite	2	110	USD	\$229.89	—	☐
NEE	Satellite	2	70	USD	\$148.89	—	☐
LOW	Satellite	2	230	USD	\$468.89	—	☐
SHW	Satellite	1	340	USD	\$348.89	—	☐
PH	Satellite	1	650	USD	\$658.89	—	☐
LIN	Satellite	1	450	USD	\$458.89	—	☐
MA	Satellite	1	480	USD	\$488.89	—	☐
ABT	Satellite	2	120	USD	\$248.89	—	☐
AVGO	High Conviction	2	230	USD	\$468.89	—	☐
MSCI	High Conviction	1	560	USD	\$568.89	—	☐
TAU	Hedge	7	55	USD	\$385.89	—	☐

Holdings table — tickers, sleeves, prices, currencies, caps.

Holdings Panel

Ticker	Stock or ETF symbol.
Sleeve	Which bucket this holding belongs to.
Shares	How many you currently own.
Price (native)	Current price in the ticker's own currency. Update before running.
Curr	Native currency — USD for US stocks, GBP for UK stocks, ILS for Israeli bonds, etc.
Value (base)	Auto-calculated: shares × price, converted to base currency.
Max %	Optional cap. Blank = no limit.
Frac	Tick if your broker allows fractional share purchases for this ticker.

4. How the Allocation Engine Works

The engine runs in three phases every time you change any input field.

Phase 1 — Sleeve-Level Allocation

For each investment sleeve, the engine calculates how much cash is needed to reach its target:

$$\text{need[sleeve]} = \text{totalAfter} \times (\text{target\%} \div 100) - \text{currentValue[sleeve]}$$

If need is negative (sleeve is over-target), it gets zero — the tool never forces selling.

Cash is distributed in strict priority order:

Priority 1 — Emergency	Top up the Cash sleeve to the 1% hard floor if it is critically low.
Priority 2 — Main	Fund all investment sleeves proportionally to their need. If total need exceeds available cash, money is split proportionally. If there is surplus, it spreads by target weight.
Priority 3 — Optional	Top up the Cash sleeve from the 1% floor to its full target %, using any cash remaining after securities are funded.

Phase 2 — Ticker-Level Allocation

Within each sleeve, the sleeve budget is split equally among eligible tickers. A ticker is excluded when it hits its Max % cap — its share overflows to the remaining tickers.

- **Frac ON:** buy the exact calculated dollar amount.
- **Frac OFF:** floor(allocation ÷ price) whole units are purchased. The sub-unit fraction is not wasted — it enters the recycle pool.

Phase 3 — Recycle Pool

After rounding, all leftover fractions within a sleeve are pooled. The engine attempts to redistribute them to any fractionable tickers in the same sleeve. If none exist, the pool becomes Cash Left.

■ *Recycled cash never crosses sleeve boundaries. Core leftovers stay in Core.*

Conservation Guarantee

$$\text{Cash Allocated} + \text{Cash Left} = \text{Available Cash to Deploy}$$

Every dollar of new cash is accounted for — either it bought securities, or it is returned as Cash Left. Fees are paid from existing cash and are not part of this identity.

5. Transaction Fees & Version B

Fee Formula

Fee per trade = Fixed Fee + (PctFee ÷ 100) × trade amount

Total fees = Σ fee for all active trades

Fee Payment Cascade

Fees are paid from these sources in order:

- 1. Cash reserve (amount set aside for the Cash sleeve floor)
- 2. Existing cash position
- 3. Stock allocation — all ticker buy amounts are scaled down proportionally, then re-rounded to whole units

■ *If fees must be raided from stock allocation, a red warning appears in the results.*

Version B — Staying Within a Fee Budget

- After initial allocation, the engine checks if total fees exceed the budget.
- If yes: the smallest trade (by dollar amount) is cut entirely.
- The freed money redistributes to the remaining tickers in the same sleeve.
- This repeats until total fees are within budget, or only one trade remains.

■ *Example: \$2.50/trade, budget \$25 = maximum 10 trades per deployment. Version B automatically trims to fit.*

6. Multi-Currency Support

The tool supports portfolios with holdings in different currencies — USD stocks, GBP-listed ETFs, ILS bonds, EUR funds.

How It Works

Base Currency	All allocation math runs in this currency. Portfolio Value, After Deploy, and Sleeve Plan are all in base.
Ticker Currency	Set the Curr field for each ticker. Price is always entered in native currency.
FX Rates	Fetches automatically from frankfurter.app. Override any rate manually in the FX Rates panel.

Conversion Formulas

Value in base = $\text{shares} \times \text{nativePrice} \div \text{FX_RATE}$

Alloc in native = $\text{allocBase} \times \text{FX_RATE}$

Units to buy = $\text{floor}(\text{allocNative} \div \text{nativePrice})$ [whole-unit]

Final buy base = $\text{units} \times \text{nativePrice} \div \text{FX_RATE}$

FX_RATE is expressed as: 1 base unit = X foreign units. Example: FX_RATE[GBP] = 0.75 means 1 USD = 0.75 GBP.

"\$ To Buy" Column in Ticker Breakdown

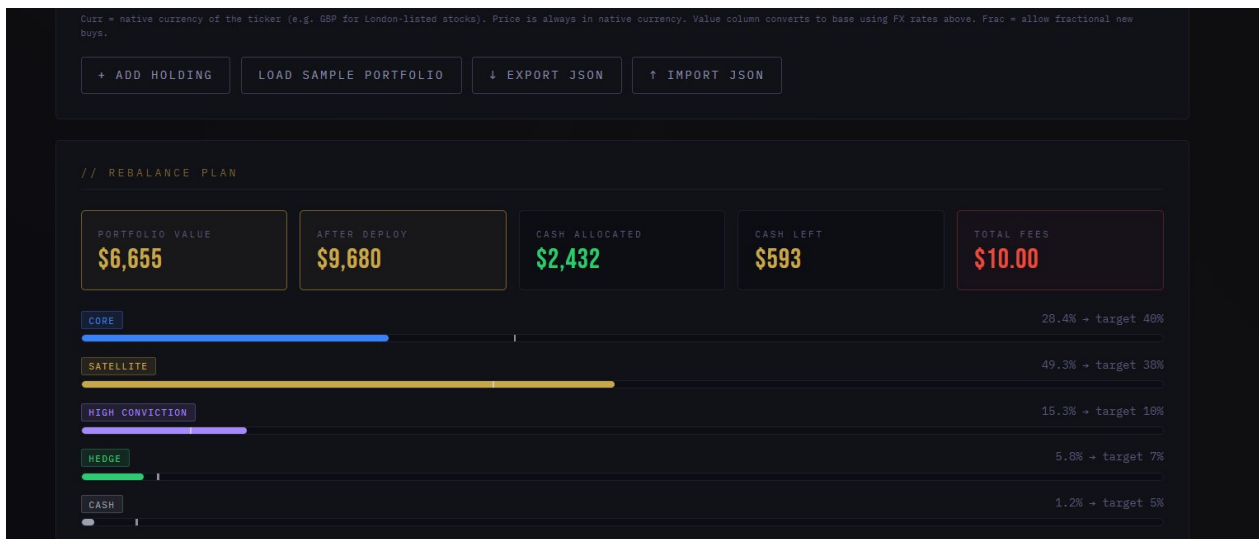
Shows the amount in the **native currency** of that ticker — this is exactly what you send to your broker. A London-listed ETF will show a GBP amount, not USD.

■ *Sleeve Plan table shows base currency amounts (for your overall plan). Ticker Breakdown shows native amounts (for executing trades).*

Missing Rates

If a rate is missing, the entire allocation is blocked until you provide it. A yellow banner shows which currencies need rates.

7. Reading the Results



Summary cards and sleeve allocation bars.

Summary Cards

Portfolio Value	totalNow — current value of all holdings + existing cash, in base currency.
After Deploy	totalAfter = totalNow + newCash. Portfolio value once all trades are executed.
Cash Allocated	Total actually spent on securities — sum of all ticker final buy amounts.
Cash Left	Unspent cash: rounding leftovers + cash reserve + overflow from capped tickers.
Total Fees	Sum of all transaction fees. Shown only when fees are configured.

Sleeve Bars

Each bar shows current allocation % (colored fill) vs target % (white marker). Fill left of marker = underweight. Fill right of marker = overweight.

SLEEVE	CURRENT %	AFTER %	TARGET %	GAP	\$ TO BUY
CORE	28.4%	40.7%	48%	+11.6%	\$2,047
SATELLITE	49.3%	35.0%	38%	-11.3%	\$110
HIGH CONVICTION	15.3%	10.5%	16%	-5.3%	—
HEDGE	5.8%	6.8%	7%	+1.2%	\$275
CASH	1.2%	1.0%	5%	+3.8%	\$19 (reserve)

TICKER	SLEEVE	CURR	CURRENT %	MAX %	UNITS	\$ TO BUY
SCHD	CORE	USD	21.6%	—	38	\$1,052.60
DGRW	CORE	USD	6.8%	—	11	\$994.29
CNI	SATELLITE	USD	4.5%	—	—	—
PLD	SATELLITE	USD	3.3%	—	1 ▲	\$110.00
NEE	SATELLITE	USD	2.1%	—	—	—
LOW	SATELLITE	USD	6.9%	—	—	—
SHW	SATELLITE	USD	5.1%	—	—	—
PH	SATELLITE	USD	9.8%	—	—	—
LIN	SATELLITE	USD	6.8%	—	—	—
MA	SATELLITE	USD	7.2%	—	—	—
ABT	SATELLITE	USD	3.6%	—	—	—
AVGO	HIGH CONVICTION	USD	6.9%	—	—	—
MSCI	HIGH CONVICTION	USD	8.4%	—	—	—
IAU	HEDGE	USD	5.8%	—	5	\$275.00

Sleeve Plan and Ticker Breakdown tables.

Sleeve Plan Table

Current %	$\text{currentValue} \div \text{totalNow} \times 100$.
After %	$(\text{currentValue} + \text{buy}) \div \text{totalAfter} \times 100$. Green = close to target. Orange = slightly off. Red = over target.
Gap	$\text{Target\%} - \text{Current\%}$. Green positive = underweight (needs buying).
\$ To Buy	Amount allocated to this sleeve in base currency.

Ticker Breakdown Table

Max %	Your cap. Red ■ = already at or above cap, receives zero allocation.
Units	Whole units to buy. Blue = normal. Purple ▲ = rounded up by overshoot tolerance.
\$ To Buy	Amount in NATIVE currency — execute this at your broker.

8. Formula Reference

Core Quantities

$$\text{totalNow} = \text{existingCash} + \sum (\text{shares}[i] \times \text{nativePrice}[i] \div \text{fxRate}[i])$$
$$\text{totalAfter} = \text{totalNow} + \text{newCash}$$

Sleeve Targets

$$\text{targetValue}[s] = \text{totalAfter} \times (\text{target\%}[s] \div 100)$$
$$\text{need}[s] = \max(0, \text{targetValue}[s] - \text{currentValue}[s])$$

Cash Floor

$$\text{cashFloor} = \text{totalAfter} \times 0.01$$
$$\text{cashReserve} = \max(0, \text{cashFloor} - \text{existingCash}) \text{ [only if existingCash} < \text{cashFloor]}$$

Ticker Allocation

$$\text{allocBase}[t] = \text{sleeveBudget} \div \text{numEligibleTickers} \text{ (equal split, cap overflow redistributed)}$$
$$\text{allocNative}[t] = \text{allocBase}[t] \times \text{fxRate}[\text{currency}]$$
$$\text{units}[t] = \text{floor}(\text{allocNative}[t] \div \text{nativePrice}[t]) \text{ [whole-unit tickers]}$$
$$\text{units}[t] = \text{allocNative}[t] \div \text{nativePrice}[t] \text{ [fractionable tickers]}$$
$$\text{finalBuy}[t] = \text{units}[t] \times \text{nativePrice}[t] \div \text{fxRate}[t] \text{ [in base currency]}$$

Fees

$$\text{fee}[t] = \text{fixedFee} + (\text{pctFee} \div 100) \times \text{finalBuy}[t]$$
$$\text{totalFees} = \sum \text{fee}[t] \text{ for all active trades}$$

Max % Cap Room

$$\text{capRoom}[t] = \text{totalAfter} \times (\text{maxPct}[t] \div 100) - \text{valueBase}[t] - \text{currentAlloc}[t]$$

If $\text{capRoom} \leq 0$, ticker is excluded.

Overshoot (Round-Up Check)

$$\text{remainder} = (\text{allocNative} \bmod \text{nativePrice}) \div \text{nativePrice}$$
$$\text{extraBase} = (\text{units}+1) \times \text{nativePrice} \div \text{fxRate} - \text{allocBase}$$
$$\text{overshoot\%} = (\text{extraBase} \div \text{totalAfter}) \times 100$$

Round up only when: $\text{remainder} \geq 0.5$ AND $\text{overshoot\%} \leq \text{tolerance}$ AND $\text{extraBase} \leq \text{capRoom}$.

Conservation Identity

$$\text{Cash Allocated} + \text{Cash Left} = \text{Available Cash to Deploy}$$

9. Common Scenarios

Scenario A — First Investment

Starting from scratch with zero holdings.

- Set each sleeve target (must sum to 100%).
- Add tickers, set current prices. Leave shares at 0.
- Enter your initial investment in "Available Cash to Deploy."
- The tool distributes cash proportionally by target weight.

Scenario B — Monthly Salary Deployment

You add a fixed amount every month. Some sleeves have drifted above target.

- Update prices (Refresh Prices with an API key, or enter manually).
- Enter this month's new cash amount.
- The tool identifies underweight sleeves and directs money there.
- Over time, the portfolio converges to target without selling.

Scenario C — Large Cash Left (All Tickers Capped)

You see a large Cash Left even though an obvious sleeve is underweight. This happens when every ticker in that sleeve is already at or above its Max % cap.

- Check the Ticker Breakdown for ■ warnings next to Max %.
- Raise the cap for tickers you want to keep buying, or add a new ticker to the sleeve.
- This is not a bug — the tool is respecting your caps.

Scenario D — Version B Fee Limiting

Your broker charges \$5 per trade. You want at most \$25 in fees (5 trades).

- Set Fixed Fee = 5, select Version B, set Max Total Fee Budget = 25.
- Version B cuts the smallest trades until total fees fit within \$25.
- Freed money from cut trades flows to the surviving tickers in the same sleeve.

Scenario E — Multi-Currency Portfolio

US stocks (USD), UK ETFs (GBP), Israeli bonds (ILS).

- Set Base Currency = USD.
 - Enter GBP or ILS in the Curr field for each foreign holding.
 - Click Refresh Rates to fetch live FX rates (or enter manually if offline).
 - All allocation math runs in USD. Ticker Breakdown shows buy amounts in native currency for broker execution.
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